

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 9: Some Applications of Trigonometry

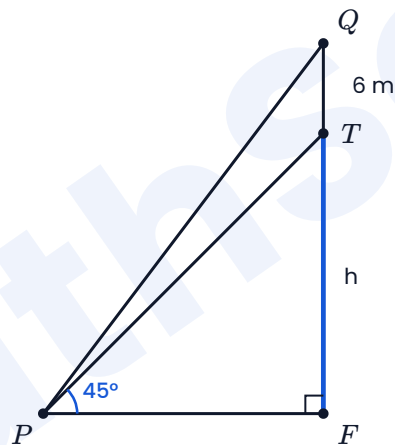
Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Two right-angled triangles

1. A 6 m pole stands on top of a tower. From a point P on the ground, the top of the pole is seen at an angle of elevation of 60° . From the top of the tower, P is seen at an angle of depression of 45° . Find the height of the tower and the distance of P from its foot. (Take $\sqrt{3} = 1.73$.)

CBSE 2024, 5 marks



FT is the tower and TQ is the pole

SOLUTION

Let the height of the tower be h m and the distance of P from its foot be d m.

The angle of depression of P from the top of the tower is 45° , so the angle of elevation of the top of the tower from P is 45° : $\tan 45^\circ = \frac{h}{d}$, so $d = h$.

For the top of the pole: $\tan 60^\circ = \frac{h+6}{d}$, so $\sqrt{3}h = h+6$.

$$h = \frac{6}{\sqrt{3}-1} = \frac{6(\sqrt{3}+1)}{2} = 3(\sqrt{3}+1) = 3 \times 2.73 = 8.19$$

Answer: the tower is 8.19 m high and P is 8.19 m from its foot.

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